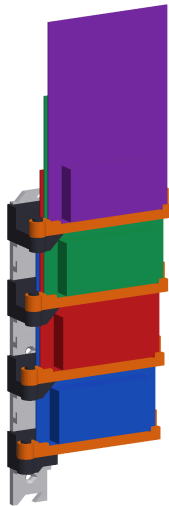


YOUR COLLECTION, ON THE WALL. EVERY CARD OPENS LIKE A DOOR.

SWINGRACK is a 3D-printed wall rack for carded die-cast cars. Each card gets its own swing-out ledge. The ledges are stacked and staggered so every name stays readable, and you can take any one card out without touching the others.



4-car kit, closed: the cards step forward 6 mm per row, so no card hides the one above it.



Early printed prototype holding real carded cars (photo by the designer).

THE PROBLEM

- Carded cars end up in boxes or on peg hooks, where they hang behind each other and the backs get bent.
- Getting one card off a crowded hook means unhooking the cards in front of it.
- Most shelf displays are built for loose cars, not for cards with a blister on the front.

THE SOLUTION

- One ledge per card, hinged on a vertical pin. Lift it 1 mm, swing it out, slide the card out.
- A slot with a low front lip and a tall back wall holds the card upright while its printed name stays visible.
- Modular: one wall strip carries up to 4 cars, and strips lock end-to-end for taller stacks.

HOW IT WORKS

01

Screw the strip to the wall

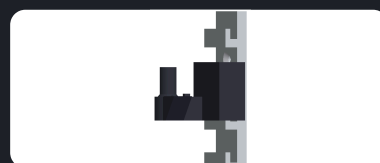
Two or three 5 mm countersunk screws. Hook lips every 20 mm.



02

Drop a clip on a hook lip

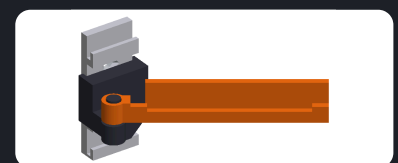
Hold it 6 mm above a lip, push it in and let go. It locks under its own weight. To remove it, lift and pull.



03

Drop the ledge on the pin

Slide a card into the slot. A small bump clicks the ledge shut and a stop nub keeps it at 0°.



3

printed part types: strip,
clip, ledge

0

support material needed to
print

2-6

car kits, more by adding
strips

~6 mm

lift and pull to take a clip
off

ENGINEERING THAT SURVIVED THE TEST PRINTS

Hooks that can't snap

The hook lips run across the full 30 mm width of the strip, and the strip prints on its edge, so the layers run along each hook. The earlier pegs broke where they met the strip.

No sagging ledge

A support arm sits under the closed ledge. A stop nub fits a notch in the ledge and holds it at 0°. A bump-and-dimple detent keeps it closed.

Stackable strips

A 45° dovetail joins 240 mm strips end-to-end, so stacks over 240 mm still screw flat to the wall.

Gravity lock, no tools

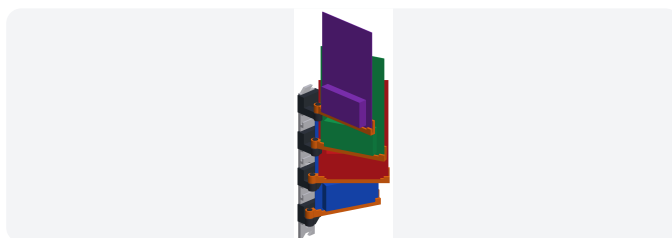
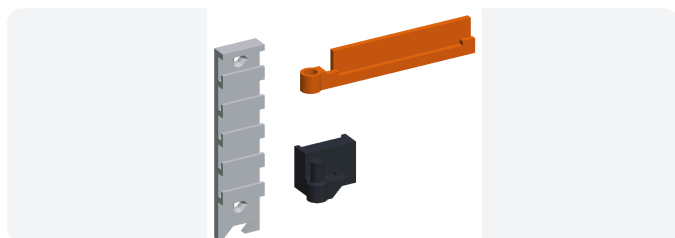
A 30 mm finger drops behind a lip. Side cheeks hug the strip so the clip can't twist. The bottom of the clip rests on the lip below it.

Names always visible

Each rack sits 6 mm further from the wall than the one below it. The ledge's low front lip leaves the card title in view.

Prints on any FDM printer

Every part was checked for overhangs: there are only two short bridges and no supports. 247 mm long parts, PLA or PETG.



KIT LINE-UP

Kit	Wall strips	Hinge clips	Ledges	Stack height	Est. filament*
2 cars	1	2 (depth 0-1)	2	119 mm	≤ 120 g
3 cars	1	3 (depth 0-2)	3	179 mm	≤ 155 g
4 cars	1	4 (depth 0-3)	4	239 mm	≤ 185 g
5 cars	2	5 (depth 0-4)	5	299 mm	≤ 280 g
6 cars	2	6 (depth 0-5)	6	359 mm	≤ 315 g

*Upper bound from STL volume printed 100% solid in PLA; real prints with normal infill use less. Fits cards about 105 × 165 mm with a blister up to 42 mm tall.

WHO IT'S FOR

- Collectors who keep their cars carded.
- Hobby and toy shops that want a neat counter or wall display.
- Makers who buy printable files on Printables, Etsy and similar marketplaces.

NEXT STEPS

- Test print the v2 hook strip and clip, then do load and drop tests with full cards.
- Launch the STL files and printed kits. Set prices after the cost test.
- Add-ons: colour sets, snap-on name plates, an 8-car kit.